

Prompt for Windsurf:

You are taking over development from another AI agent on a project called **PulseCare Telehealth Scheduler**, a mid-complexity healthcare application built with:

- **Backend:** Java 21, Spring Boot 3 microservices, Maven, Spring Cloud (Eureka, Config, Gateway)
- **Frontend:** React (Staff Portal) + Vue.js (Patient Portal)
- **Databases:** PostgreSQL for transactional data, MongoDB for audit logs
- **Deployment:** Docker, Kubernetes, GCP (GCR + GKE), GitHub Actions CI/CD
- **Scaling:** Horizontal Pod Autoscaler (HPA)
- **Monitoring:** Prometheus + Grafana
- **HIPAA-friendly:** Role-based JWT auth, PII masking
- **Target:** PST/EST work hours alignment

Current state:

- Repo: <https://github.com/sachinschitre/pulsecare-telehealth>
- Directory structure, many backend services, some Docker and K8s manifests are created.
- .env file is populated with GCP params.
- GitHub Actions secrets set.
- GCP project created.
- Some scripts (`make init`, GCP deploy scripts) fail because of missing files/configs.
- Frontend code (React & Vue) is not yet implemented — only folder placeholders exist.
- Local testing via `docker-compose` fails because of service misconfigurations and missing dependencies.

Your task:

1. Audit & Fix Missing Pieces

- Identify and create any missing files, configs, Dockerfiles, and Kubernetes manifests so the entire system builds and runs without error.
- Ensure each microservice has:
 - `pom.xml` with correct dependencies.
 - `application.yml` / `application.properties`.
 - Dockerfile with multi-stage build (distroless or Alpine base).
 - Working Spring Boot entrypoint.
- Ensure all Docker services build and start with `make up`.

2. Frontend Implementation

- Build **React Staff Portal**:
 - Login (Admin/Provider) with JWT auth.
 - Provider availability management.
 - Appointment management dashboard.
 - Real-time telehealth session launch (Jitsi).
- Build **Vue Patient Portal**:
 - Login/Register as Patient.
 - Search providers by specialty & availability.
 - Book/cancel appointment.
 - Join telehealth session.
- Use **sleek, modern UI** (TailwindCSS or Material UI).

- Implement API integration with backend gateway.

3. Local Development (Docker)

- Make `docker-compose.yml` fully functional for local dev with:
 - PostgreSQL + MongoDB init scripts.
 - All backend services.
 - Both frontends.
 - API Gateway.
 - Config & Discovery services.
- Ensure `make init` installs all dependencies, `make up` runs full stack, and `./scripts/local_seed.sh` seeds demo data.

4. Testing

- Add JUnit 5 + Testcontainers for backend integration tests.
- Add Playwright or Cypress E2E tests for frontends.
- Ensure `make test` runs all tests locally and in CI.

5. Deployment to GCP

- Fix and complete scripts for:
 - Building & pushing images to **Google Container Registry**.
 - Deploying all services to **Google Kubernetes Engine**.
 - Applying Kubernetes manifests (`kubectl apply -f ...`).
 - Setting up Ingress + SSL (self-signed if needed for demo).
- After deploy, app should be accessible via public URL and HPA scaling demonstrable.

6. Metrics & Monitoring

- Deploy Prometheus + Grafana to GKE.
- Provide a sample Grafana dashboard JSON for monitoring CPU, memory, request count, and HPA pod count.

7. Documentation

- Update `README.md` with:
 - Project overview.
 - Local dev instructions.
 - Testing instructions.
 - GCP deploy guide.
 - Demo script for interview presentation.

8. Final Output

- Fully working app locally and on GCP.
- Frontend + backend integrated.
- CI/CD passing.
- Demo-ready in 1–2 hours of your work.

Efficiency Metrics Requirement (for Interview Presentation):

- Include in the final README a “**Time & Efficiency Gains**” section that:
 - States that **manual development of this stack with one junior dev would take ~16 hours (2 full days).**
 - States that using AI-assisted dev + strong human direction reduced it to **~2 hours, an 88% time savings.**

- Break down the time saved by stage (architecture setup, backend build, frontend build, deployment).
- Present the result as:
 - **Original Estimate:** 16 hours
 - **Actual Delivery:** 2 hours
 - **Productivity Multiplier:** 8× faster delivery
 - **Efficiency Gain:** 88% reduction in dev time

Important Constraints:

- You should optimize for speed: reuse existing code where possible, don't over-engineer.
- Use secure coding & HIPAA-friendly practices.
- Output all missing or modified files directly in this workspace.
- Run `make up` and `make test` locally to verify before final GCP deploy.